Progress Report 2

Weeks 1 and 2: set-up, data verification and the preprocessing pipeline

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Project: LipScribe – Lip-Assisted Speech Recognition for Robust Live Captioning 29 October 2026

1 Summary

In the first two weeks (19–29 October) the team completed all of Week 1 and most of Week 2. The GRID corpus is downloaded and verified on the team's GPU workstation, split by speaker, and feeds a tested pipeline that turns each utterance into mouth crops, 16 kHz audio mixed with noise and a CTC target, batched and ready for training. The first model component, the audio encoder, is finished and tested, so every part of the audio-only baseline (B1) now exists. Milestone M1 is closed; Milestone M2 (due Sunday 1 November) is open because the full mouth-crop extraction (W2-02) still has to run over all 33 usable speakers and B1 has not yet started training. The figures in this report come from the repository, its test suite and the integrity report produced on the workstation, not from the Trello board alone.

22 / 26
task cards done (Week 1: 12 of 12, Week 2: 10 of 14)

96.7 %
of 34,000 utterances pass every integrity check

160 / 163
tests pass on *main*; the rest need workstation clips

25 / 28
merged pull requests have a recorded review

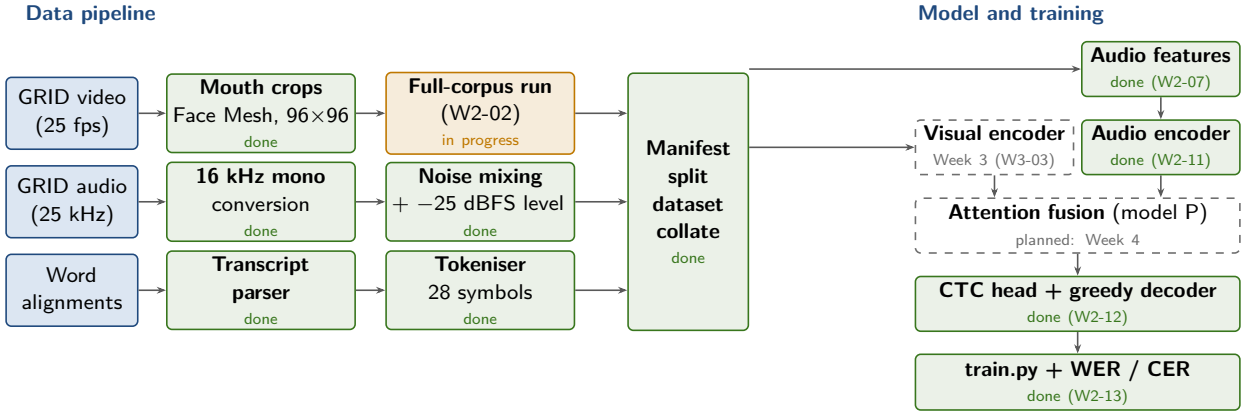


Figure 1: The LipScribe pipeline and the status of each stage on 29 October. All data stages are built and tested except the full-corpus crop run; every part of B1 (audio encoder → CTC head) exists, and the visual encoder and fusion follow in Weeks 3–4.

2 Progress against the plan

Milestone	Due	Status
M1 Environment and dataset verified	Sun 25 Oct	**Done** – closed 29 Oct, 4 days late
M2 Preprocessing and alignment	Sun 1 Nov	**In progress** – 3 of 6 criteria met
M3 & M4 Baselines B1, B2, B3	Sun 8 Nov	Planned
M5 Multimodal fusion (model P)	Sun 15 Nov	Planned
M6 & M7 Tuning and evaluation (core freeze)	Sun 22 Nov	Planned
M8 & M9 Application, testing, demo	Sun 29 Nov	Planned

M2 criteria met: crop contact sheets reviewed, noise-mixing tests pass, and one batch loads with consistent lengths. Still to do: the train/inference parity test, a running B1 job and the Week 3 re-plan. Most Trello cards were moved to *Done* in one update on 29 October, so the pull-request history (Section 5) is the more reliable record of when work finished.

3 Week 1: set-up and data (Milestone M1)

Card	Task	Owner	Output and evidence
W1-01	Kick-off, roles, Trello	Charis	Board: every card has an owner, reviewer, due date, checklist
W1-02	Repository skeleton	Nehum	Package `lipscribe/`, one sub-package per stage (PR #2)
W1-03	Pinned environment	Nehum	`requirements.txt`, `environment.yml`, `check_env.py`
W1-04	GPU and storage	Charis	Linux GPU workstation (“Alienware”) runs all full-corpus jobs
W1-05	Download GRID	Faith	34 speakers × 1,000 utterances: video, audio, alignments
W1-06	MUSAN noise pools	Lisa	Split *by file* 70/15/15: test noise never heard in training
W1-07	Integrity checker	Faith	`verify_dataset.py`: 7 checks per utterance (Table 1)
W1-08	Speaker split	Faith	`make_splits.py`: fixed seed, refuses to overwrite
W1-09	Tokeniser	Lisa	28 symbols; verified on all 33,000 alignment files
W1-10	Mouth-crop prototype	Cole	Face Mesh lip crops, 96×96 grey; tested on five clips
W1-11	Interface specification	Isabelle	`INTERFACES.md`: every function, key and shape (PR #4)
W1-12	Tests and CI	Nehum	pytest suite and a GitHub Actions file (PR #6)

`docs/INTERFACES.md` fixed every module’s inputs, outputs and tensor shapes before the modules were written, so the data, model and application pairs could build in parallel; the Week 2 integration relied on it directly.

3.1 Dataset integrity

`verify_dataset.py` ran over all 34,000 utterances on the workstation. Table 1 gives the pass count for each check. Almost every failure comes from two speakers: s21 is distributed with audio only (no video or alignments), and 108 of s8’s videos do not decode. Beyond these, one s4 video does not decode and one s28 clip fails the audio-level check. The Face Mesh check (at least 90 % of frames detected) is built in and is fed by the crop extraction of W2-02.

The report also revealed a property of this copy of GRID that the pipeline must handle: the audio is *end-pointed*. Each file holds only the spoken part, 1.10–2.94 s long (mean 1.83 s), while every video lasts 3.0 s, and speech starts on average 0.47 s (at most 1.51 s) into the video. All files are pre-levelled to −26 dBFS. Without correction the audio would run ahead of the lips; Section 4.7 describes the fix.

Table 1: Integrity check results (`reports/integrity_report.csv`, 34,000 utterances).

Check (per utterance)	Pass	Rate	Notes
Video, audio and alignment files present	33,000	97.1 %	s21: no video or alignment
Video decodes	32,891	96.7 %	s8: 108 · s4: 1
Frame count 75 ± 2	32,891	96.7 %	32,816 at 75 frames, 74 at 74
Audio–video sync (≤ 40 ms)	32,891	96.7 %	20 ms offset on every clip
Transcript is six valid GRID words	33,000	97.1 %	51 words; 20–31 characters
Audio not silent or clipped	33,999	100.0 %	s28: 1 clip
All checks	**32,890**	**96.7 %**	33 of 34 speakers usable

3.2 Speaker-independent split

Speakers, not utterances, are split, so no test speaker is ever heard or seen in training (Figure 2). The test set has two female and two male speakers and the validation set three speakers of both sexes; every other usable speaker trains. The split was generated once with a fixed seed (2026) and is committed; `make_splits.py` refuses to overwrite it, and a test asserts that the three speaker sets are pairwise disjoint.

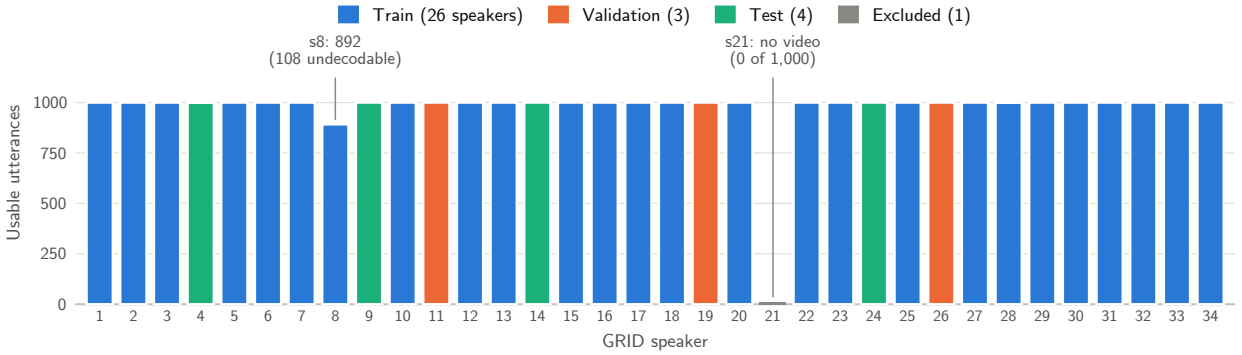


Figure 2: Usable utterances per GRID speaker after the integrity checks, coloured by partition. Train: 26 speakers (14 male, 12 female), 25,891 utterances. Validation: s11, s19, s26 (2 male, 1 female), 3,000. Test: s4, s9, s14, s24 (2 female, 2 male), 3,999. Excluded: s21.

3.3 Transcripts and tokeniser

Transcripts come from the `.align` files with silence (`sil`) and short pauses (`sp`) removed. The tokeniser maps them to 28 symbols: blank (0), space (1) and the letters a–z (2–27). Lisa's `verify_tokenizer.py` checked it on all 33,000 real alignment files: $\mathit{decode}(\mathit{encode}(s)) = s$ for every transcript, the blank is never emitted, and the vocabulary holds the 51 GRID words. The longest transcript is 31 characters ("place green with e eight please") and the most repeated adjacent letters in one transcript is 3. CTC therefore needs at most $31 + 3 = 34$ output frames against 75 available per clip, so the CTC length condition can never fail on GRID.

4 Week 2: preprocessing and training pipeline (Milestone M2)

Ten of the fourteen Week 2 cards are done (Table 2); the four open ones all wait on the full crop extraction or on B1 training.

4.1 Mouth crops

MediaPipe Face Mesh finds the lip landmarks in every frame. A square box around them, enlarged by 25 %, is smoothed with a centred five-frame moving average and resized to 96×96 grey pixels. When detection fails, the box is interpolated from the nearest good frames and the frame is marked invalid in a per-frame mask; frames are never dropped, so the video stays aligned with the audio.

Table 2: Week 2 cards and their status on 29 October.

Card	Task	Owner	Status	Output and evidence
W2-01	Crop failure handling	Cole	**Done**	Failed frames interpolated, never dropped
W2-02	Full mouth extraction	Faith	**In progress**	Script merged; s1 pilot run done
W2-03	Crop contact sheets	Isabelle	**Done**	Six speakers reviewed (Figure 3a)
W2-04	Audio to 16 kHz	Lisa	**Done**	34,000 of 34,000 files, 0 failures
W2-05	Noise mixer	Lisa	**Done**	SNR within 0.01 dB of target (PR #22)
W2-06	Fixed noisy val/test sets	Lisa	**Done**	`make_noisy_test.py` (PR #22)
W2-07	Feature extraction	Charis	**Done**	Matches librosa to 4.9×10^{-5}
W2-08	Normalisation statistics	Faith	Not started	Needs the crops from W2-02
W2-09	Manifest, dataset, collate	Charis	**Done**	Batch of 16 loads (PR #18, #21)
W2-10	Parity test	Nehum	**In progress**	Needs the crops from W2-02
W2-11	Audio encoder	Cole	**Done**	5 of 5 tests pass (PR #27)
W2-12	CTC head, decoder, metrics	Isabelle	**Done**	Corpus-level WER/CER (PR #13)
W2-13	Training script	Charis	**Done**	Trains, saves, resumes and logs
W2-14	GPU memory, B1 overfit	Charis	Planned	Unblocked by W2-11

A clip with more than 10 % invalid frames is excluded (Figure 3b). On the speaker-1 pilot run (1,000 clips) every clip was cropped without error and 984 had all 75 frames detected; the average failure rate was 0.19 % of frames, and 11 clips (1.1 %) exceeded the 10 % limit. The crop review (W2-03) produced contact sheets for six speakers (s1, s2, s10, s20, s30, s34; 140 sampled frames, all detected) and kept the 25 % margin and five-frame smoothing unchanged. `extract_mouths.py` runs the same code over the whole corpus in one command, in parallel across all CPU cores, and can be stopped and restarted without losing work; it writes the Face Mesh report that `verify_dataset.py` reads.



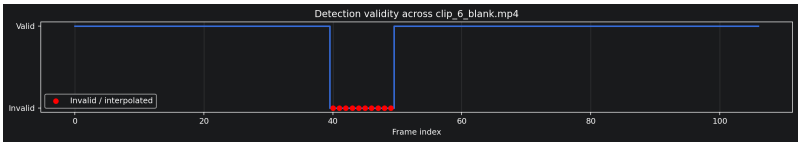


Figure 3: (a, left) Contact sheet for speaker s20: frames 0, 25, 50 and 74 of five utterances, all detected and centred on the mouth. (b, above) Failure handling on a test clip with frames 40–49 blanked out: those frames are marked invalid and their crop box is interpolated from the neighbouring frames, while the clip keeps every frame.

4.2 Audio and noise

All GRID audio is converted to 16 kHz mono, 16-bit WAV (34,000 files, no failures). Noise from the MUSAN pools is added at a chosen signal-to-noise ratio by scaling it by $\alpha = \sqrt{P_s/(P_w \cdot 10^{\mathrm{SNR}/10})}$, where P_s and P_w are the speech and noise powers over the whole clip; the measured SNR is checked to within 0.01 dB, short noise is looped, and a mixture that would clip is scaled down as a whole, which preserves the SNR. Training draws an SNR uniformly from {clean, 10, 5, 0, −5} dB for every utterance; validation and test use fixed pre-mixed files so every model hears identical audio. After mixing each clip is levelled to −25 dBFS, with the gain capped at +20 dB.

4.3 Audio features

One shared module computes the features for both training (on the GPU, in batches) and the demo app (one clip at a time). The configuration switch `audio_features` selects one of three feature sets (Table 3). All use a 25 ms Hann window, 10 ms hop, 512-point FFT and 80 Mel bands between 20 Hz and 8 kHz, with $\log(x + 10^{-6})$, and give exactly four feature frames per video frame (300 for a 3 s clip). The log-Mel output matches `librosa` to 4.9×10^{-5}.

Table 3: Audio feature configurations.

ID	Config value	Features	Shape (3 s)
A1	`logmel80`	80 log-Mel bands (default; used by B1, B3, P)	[300, 80]
A2	`mfcc39`	13 MFCCs with first and second differences (Δ, ΔΔ)	[300, 39]
A3	`logmel_mfcc`	A1 and A2 concatenated (experiment X5)	[300, 119]

4.4 Audio encoder

The audio encoder (W2-11) turns standardised log-Mel features into one 512-dimensional vector per video frame (Table 4). Two stride-2 convolutions reduce the 100 frames per second of the features to the 25 per second of the video, so audio and video can later be fused frame by frame; a two-layer bidirectional LSTM then models the sequence. Padded frames are excluded with packed sequences, and the true output lengths, $\lceil T_a/4 \rceil$, agree with the lengths the collate function computes. Its five tests cover the length formula, the output shape with and without explicit lengths, that a short clip gives the same output alone as inside a padded batch (within 10^{-5}), and that gradients flow. Joined to the CTC head, the B1 model has 3.55 M parameters, and a batch of sixteen 3 s clips gives CTC log-probabilities of shape [75, 16, 28], as `train.py` expects.

Table 4: Audio encoder layers and the CTC head for B1 (batch B, 3 s clip).

Layer	Output shape	Parameters
Input: standardised log-Mel features	$[B, 300, 80]$	–
Conv1d 80→256, kernel 3, stride 2 + BatchNorm + ReLU	$[B, 150, 256]$	62,208
Conv1d 256→256, kernel 3, stride 2 + BatchNorm + ReLU	$[B, 75, 256]$	197,376
BiLSTM, 2 layers, 256 units per direction, dropout 0.3	$[B, 75, 512]$	2,629,632
Audio encoder total		**2,889,216**
CTC head: BiLSTM 128 per direction → Linear 256→28 → log-softmax	$[75, B, 28]$	664,604
B1 total		**3,553,820**

4.5 Data loading and the training script

`build_manifest.py` joins the integrity report, crops, 16 kHz audio and split into one row per utterance, with a reason for every unusable clip. The dataset and collate function return the batch format fixed in `INTERFACES.md`: video $[B, 1, T, 96, 96]$ scaled to 0–1, padded waveforms (features are computed on the GPU), feature and encoder lengths, a padding mask $[B, T]$ and concatenated CTC targets; they reject any clip that breaks the CTC length condition. `train.py` reads one YAML configuration and implements Section 8 of the implementation plan (Table 5). Every run gets its own directory with the configuration and Git commit, saves the latest and the best checkpoint (by mean validation WER), logs to TensorBoard, can resume, and appends one row to `experiments/results.csv`.

Table 5: Training settings in `train.py` (starting values, to be tuned on validation).

Setting	Value
Optimiser	AdamW, learning rate 3×10^{-4}, weight decay 10^{-2} (none on biases and norms)
Schedule	one epoch of linear warm-up, then cosine decay
Batch	16 utterances; gradient accumulation available if memory is short
Stability	mixed precision; gradients unscaled, then clipped at norm 5.0; LSTM forget-gate bias 1
Loss	CTC (blank 0, infinite losses zeroed); non-finite batches skipped and logged
Validation	loss and greedy WER/CER at every SNR and their mean, after each epoch
Stopping	early stopping after 8 epochs without improvement; at most 60 epochs

4.6 Problems found and fixed during integration

Connecting the modules end to end exposed five faults that unit tests on each module alone had missed. All are fixed on `main` and covered by tests:

- **Transcripts.** The alignment parser read the time column instead of the word, so every transcript would have been numbers.
- **Audio–video sync.** End-pointed audio started at frame 0 while the lips start about 0.47 s later; the dataset now shifts each trimmed clip by its speech start time from the alignment (PR #21).
- **Noise variety.** Every data-loading worker drew the same SNR sequence, repeated every epoch; each worker is now reseeded per epoch, still reproducible from the run seed (PR #21).
- **Loudness.** The −25 dBFS levelling was not applied after mixing, so loudness varied with SNR (PR #21).
- **Git.** An ignore rule meant for the face-model file also hid `lipscribe/models/`, so model code would never have been committed.

5 Quality and process

Tests. The suite has 163 automated tests in 16 files: tokeniser and transcripts 64, dataset and collate 15, noise mixer 12, audio features 12, mouth cropping 10, integrity checker 7, metrics 7, training script 6, speaker split 6, decoding 5, audio encoder 5, and 14 for the noisy sets, extraction, CTC head and set-up. On `main` 160 pass and 1 is skipped; the 2 remaining crop tests need video clips that exist only on the workstation.

Code review. 28 pull requests are merged, 25 with a recorded review from the card's reviewer; the other three merged `main` into a working branch. The repository has 94 commits and about 5,600 lines of Python in 47 files.

CI and records. The workflow file is in `.github/workflow/`, but GitHub only runs `.github/workflows/`, so CI has never run. Meeting notes and `docs/compute.md` (GPU and booking rules) are not yet written.

5.1 Contributions by member

Member	Role	Cards done (Weeks 1–2)	Open
Charis Opol	Coordinator, training	W1-01, 04; W2-07, 09, 13	W2-14
Cole Rodney Migisha	Model lead	W1-10; W2-01, 11	–
Ainomugisha Faith	Data lead	W1-05, 07, 08	W2-02, 08
Lisa Lusinga Namugenyi	Audio and evaluation	W1-06, 09; W2-04, 05, 06	–
Wakisa Isabelle	Application lead	W1-11; W2-03, 12	–
Ansasire Nehum Naamaye	QA and live systems	W1-02, 03, 12	W2-10

Card owners as recorded on Trello. Review pairs: Faith–Lisa, Cole–Charis, Isabelle–Nehum.

6 Risks and next steps

6.1 Risks

1. **W2-02 is the critical path.** The full extraction (about 23 GB, several hours) blocks the normalisation statistics (W2-08), the parity test (W2-10) and every model that uses video (B2, B3, P).
2. **No training results yet.** B1 is fully built but has not trained, so there are no GPU memory figures, epoch times or WER/CER values yet, and Week 3's baselines all start from B1.
3. **Untested at full scale.** The pipeline has run end to end on small synthetic data, not yet on the workstation with all 32,890 clips, where memory and data-loading speed may differ.
4. **Repository size.** The 33,000 alignment files, their ZIP archive and a 44 MB project archive (`cv_group_4.zip`) were committed to Git, against the working agreement that no data goes into the repository.

6.2 Next steps (to Sunday 1 November)

1. Cole and Charis: add the small `build_model` function that joins the audio encoder to the CTC head for `train.py`.
2. Charis and Nehum: measure GPU memory at batch size 16, check that B1 overfits 8 utterances (CTC loss below 0.05, CER 0 within 300 steps), record the epoch time and launch the full B1 run (W2-14); record the GPU in `docs/compute.md`.
3. Faith: run `python -m scripts.extract_mouths` on the workstation, re-run `verify_dataset.py` with the Face Mesh report and rebuild the manifest; then compute the normalisation statistics (W2-08) with Lisa.
4. Nehum: rename the CI folder so tests run on every pull request, and finish the parity test (W2-10).
5. All: hold the M2 review on 1 November and move unfinished cards into Week 3 with a reason. If the schedule slips, optional extensions go first; B1–B3, P and per-SNR evaluation are never dropped.

6.3 Outlook for Weeks 3–6

Week	Focus	Main work	Milestone
3	Baselines	Evaluate B1; visual encoder; train B2 and B3; beam search	M3/M4, 8 Nov
4	Fusion	Cross-modal attention, gating, modality dropout; model P	M5, 15 Nov
5	Tuning, evaluation	Sweeps, final runs, results at every SNR; live captions	M6/M7, 22 Nov
6	Delivery	Tests, README, final report, slides, demo video	M9, 29 Nov